

Thinh Nguyen

San Jose, CA | 206-581-7487 | mthinhngn@gmail.com | linkedin.com/in/mthinhngn | github.com/mthinhngn

EDUCATION

San Jose State University

San Jose, CA

Bachelor of Science in Computer Engineering

Expected: December 2027

- **Organizations:** SJSU Software and Computer Engineering Society (Development Team), Spartan Racing
- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Computer Architecture, Software Engineering I

EXPERIENCE

SJSU Computer Science Systems Group

August 2026 – Present

Undergraduate Research Assistant

San Jose, CA

- Collaborated with 10+ researchers on DDD, delivering internet services through distributed Android infrastructure
- Analyzing encrypted client-transport-server exchange with Java/Kotlin, Spring Boot, gRPC, and Wi-Fi Direct
- Evaluated replay protection and data-freshness checks, identifying reliability risks across intermittent networks

SJSU Software and Computer Engineering Society

January 2025 – Present

Development Team Officer

San Jose, CA

- Built automated deployment and monitoring platform with FastAPI, Docker, and Prometheus for 3,000+ users
- Engineered GitOps workflows with GitHub/Smee and YAML configs, shortening deployment time by 80%
- Configured Prometheus monitoring across 15+ services for real-time health, availability, and performance tracking
- Developed a public status page showing real-time health and 90-day history using Python requests and dataclasses
- Automated Discord alerts with Git/Docker logs and Nginx caching, reducing server traffic by 60%

SJSU College of Engineering

May 2026 – August 2026

Software Engineering Intern

San Jose, CA

- Built an Express.js API streaming YouTube playlists to Raspberry Pi speakers for 1,000+ users in club community
- Integrated the YouTube API with retry logic and error handling, achieving 95% uninterrupted playback
- Implemented real-time monitoring with Docker, Prometheus, and Grafana, reducing system downtime by 20%

Spartan Racing

August 2025 – May 2026

Software Engineering Intern

San Jose, CA

- Engineered a software-in-the-loop (SIL) framework in pytest with mock I/O drivers to validate VCU algorithms
- Automated tests via GitHub Actions CI/CD, cutting turnaround time 60% and minimizing hardware debugging
- Extended SIL to hardware-in-the-loop (HIL), catching ECU timing faults undetectable in simulation
- Validated CAN bus communication with BusMaster and i2 Pro, verifying message timing and signal scaling

PROJECTS

Porta | *FastAPI, OpenAI, Ollama, Docker, Kubernetes, Helm, Redis, Prometheus, Grafana*

- Built an LLM gateway with automatic failover and caching to keep AI apps running across providers and cut costs
- Designed OpenAI to Ollama fallback routing across Llama and Qwen, sustaining 93% success on primary failure
- Implemented Prometheus metrics and Grafana dashboards for token usage and fallback request rates

PreParty Live | *Swift, Golang, PostgreSQL, AWS RDS, S3, CloudFront*

- Developed a real-time pre-event engagement app with SwiftUI and Go, enabling live chat and interactive activities
- Implemented WebSockets for low-latency, bidirectional messaging between SwiftUI clients and the Go backend
- Deployed on AWS ECS with PostgreSQL RDS, S3, and CloudFront for scalable storage and media delivery

Adversarial Text Generator | *Python, Hugging Face, PEFT, TRL, PyTorch, QLoRA*

- Fine-tuned Llama 3.2 1B with QLoRA on HateXplain, generating adversarial text for classifier training
- Built a six-metric suite evaluating toxicity, diversity, fluency, attack success, control accuracy, and LLM-as-judge
- Developed a Hugging Face and PyTorch pipeline for Colab training, ablation studies, and decoding analysis

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript/TypeScript, HTML/CSS, Bash, Go (Golang)

Frameworks: React, FastAPI, Node.js, Next.js, Express.js, Flask, JUnit

Databases & Storage: Postgres, MongoDB, Oracle, Redis, S3

Infra & Tools: Docker, Kubernetes, Terraform, Linux, GCP, Git, AWS, CI/CD, Prometheus, Grafana